



Swiss Institute of
Bioinformatics

DAY 2, PART 2

Structure comparison metrics and confidence metrics

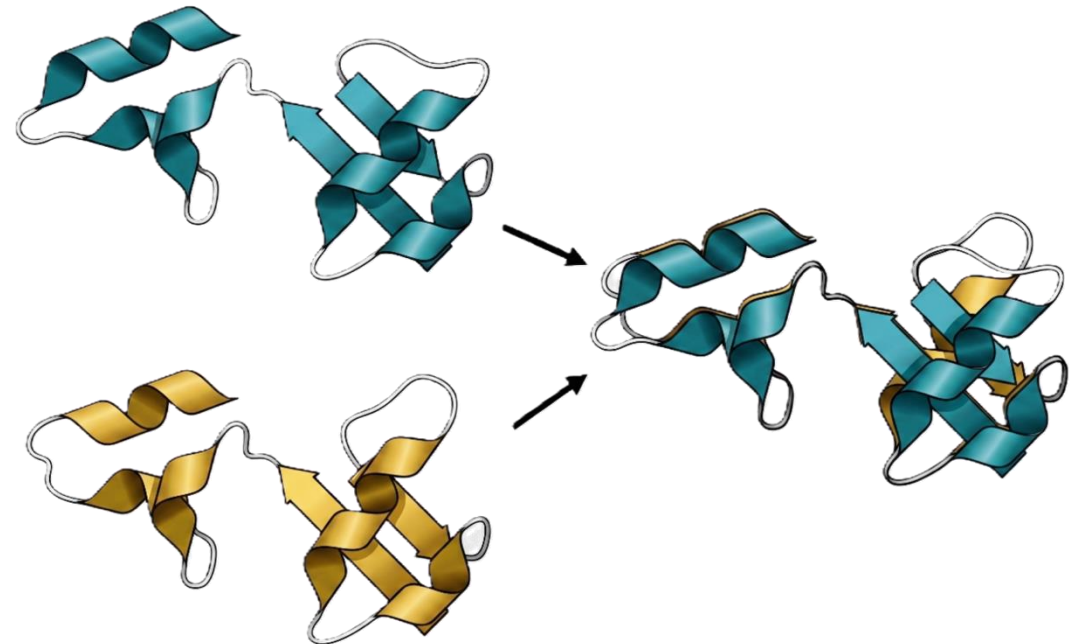
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23-24 June 2026



Structure comparison

- To determine prediction quality
 - Prediction vs experimental ground truth
- To determine structural diversity
 - Conformation 1 vs conformation 2

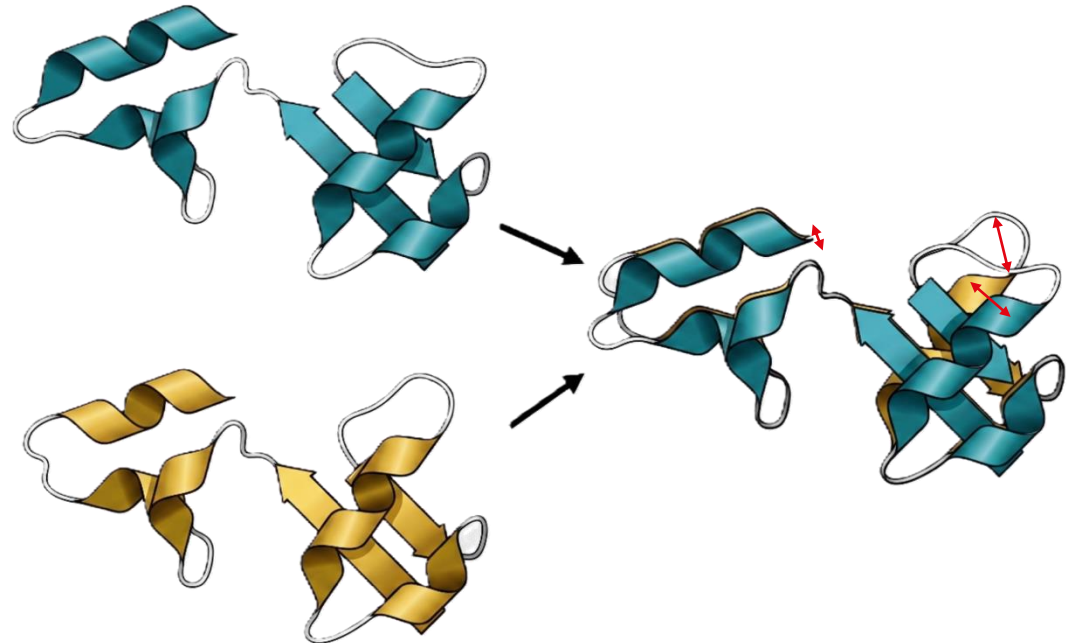


RMSD (Root mean square deviation)

- Needs structural alignment
- Distances between atoms in structure 1 and structure 2

$$RMSD = \sqrt{\frac{1}{N} \sum_{i=1}^N d_i^2}$$

- Size dependent
- Lower is better

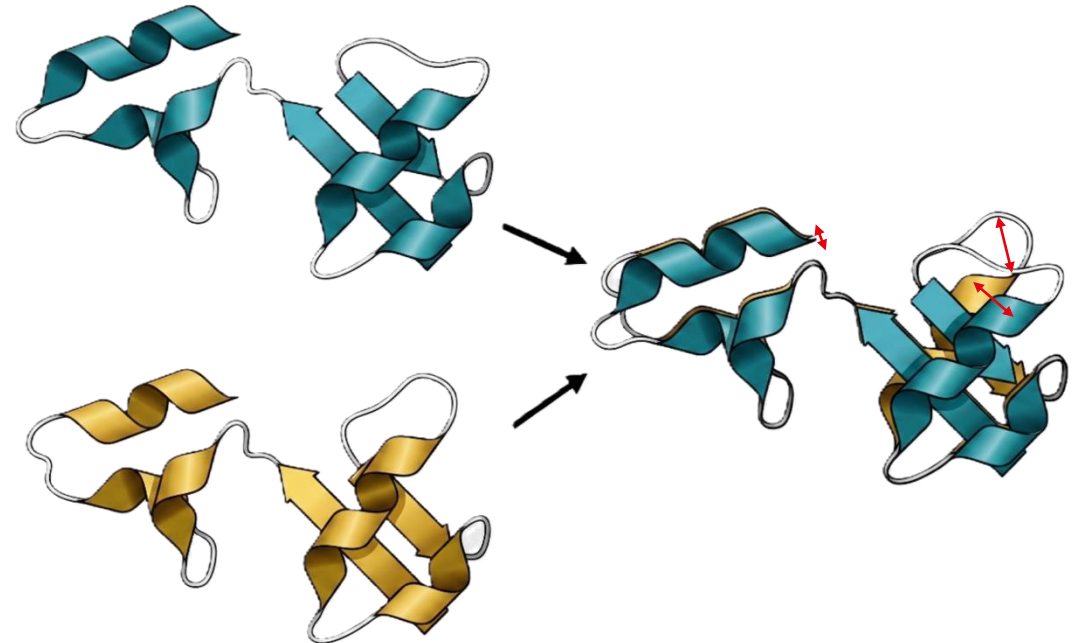


TM (template modelling) score

- Needs structural alignment
- Distances between atoms in structure 1 and structure 2

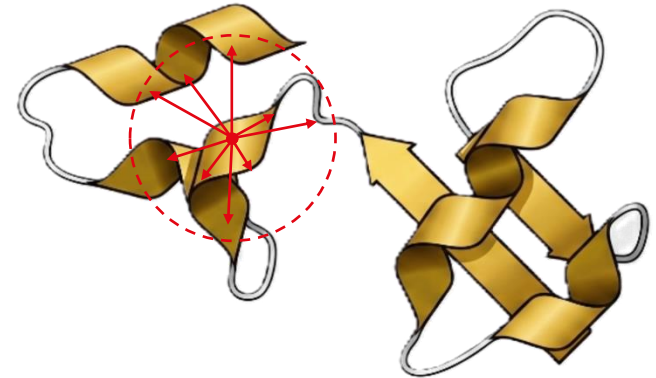
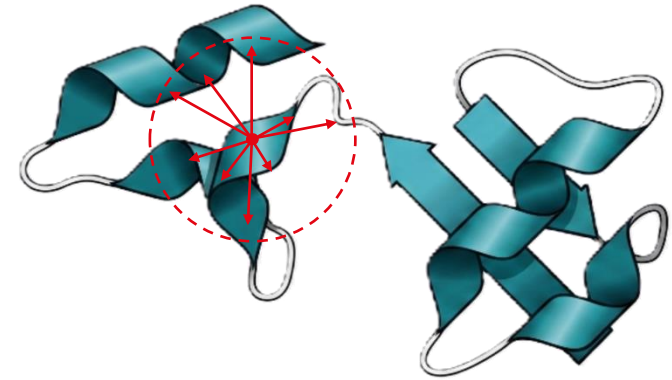
$$TM = \max \left(\frac{1}{L_N} \sum_{i=1}^{L_T} \frac{1}{1 + \left(\frac{d_i}{d_0} \right)^2} \right)$$
$$d_0 = 1.24 \times \sqrt[3]{L_N - 15} - 1.8$$

- Size independent
- Range: 0-1
Higher is better



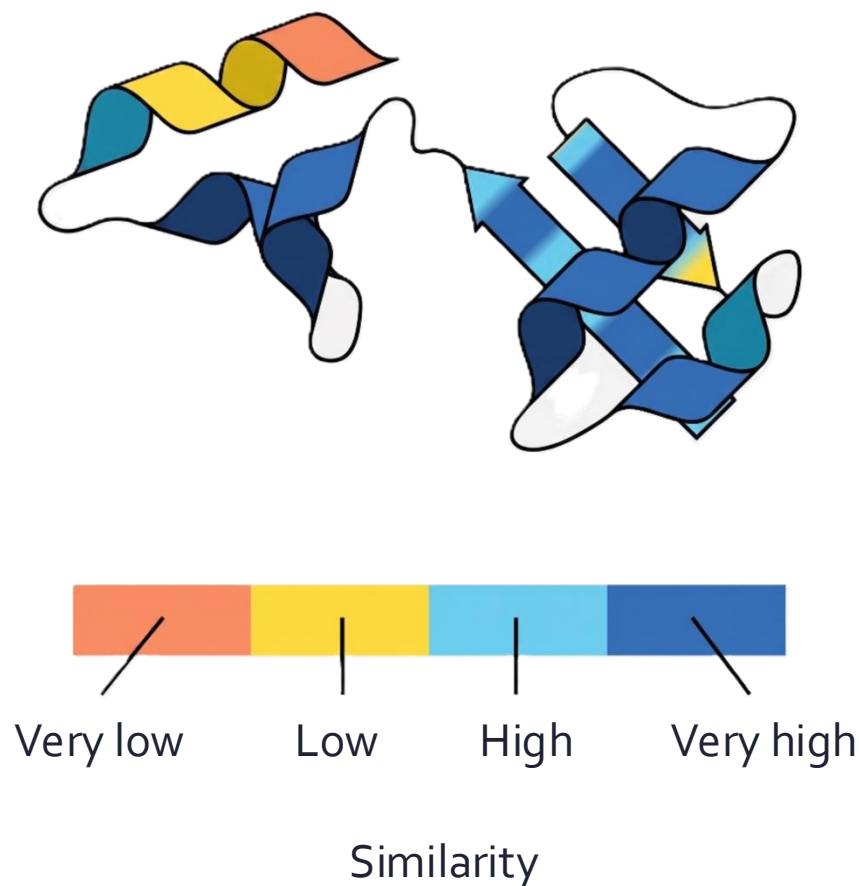
LDDT (local distance difference test)

- Local metric, per residue
- Does not need alignment
- Fraction of conserved distances within a cutoff radius
- Range: 0-100
Higher is better



LDDT (local distance difference test)

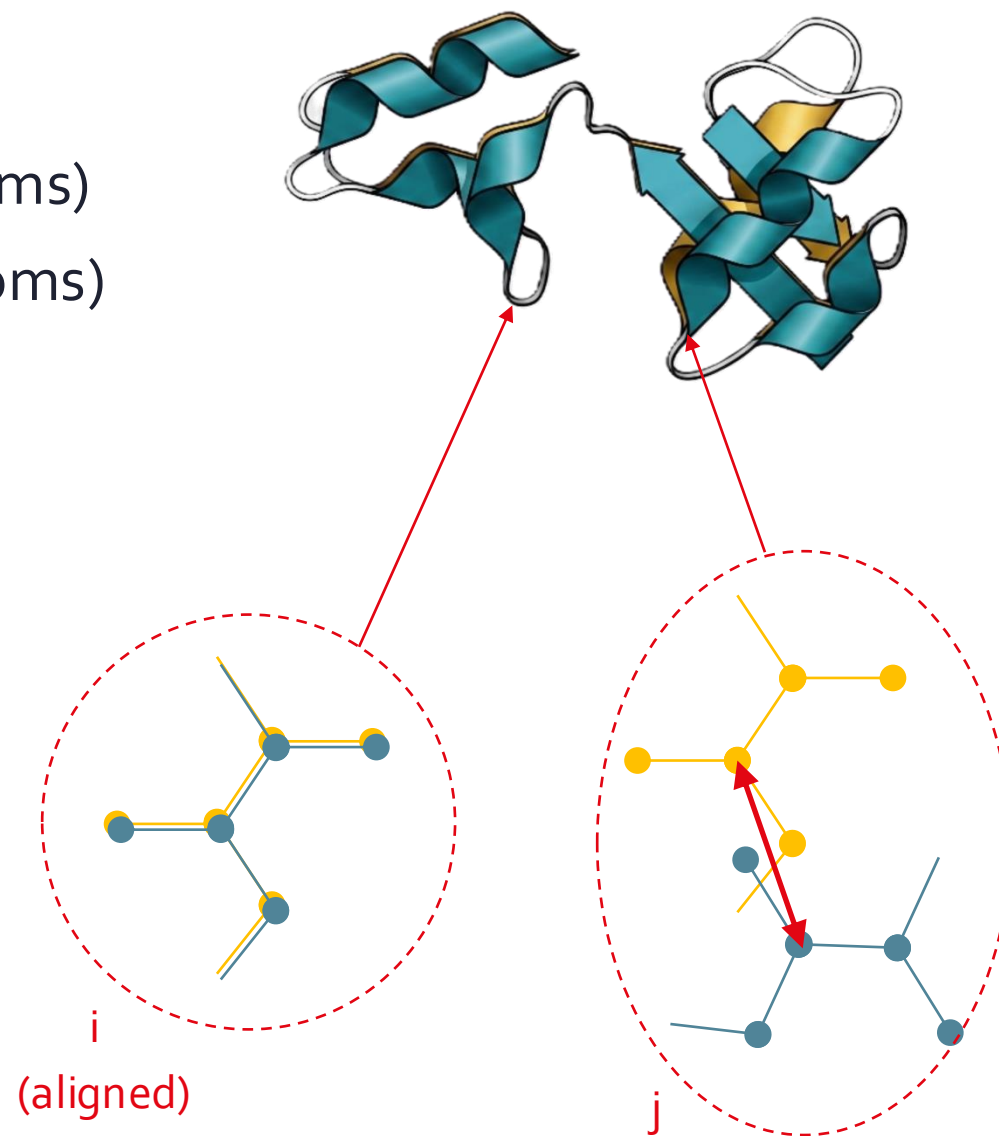
- Local metric, per residue
- Does not need alignment
- Fraction of conserved distances within a cutoff radius
- Range: 0-100
Higher is better





AE (aligned error)

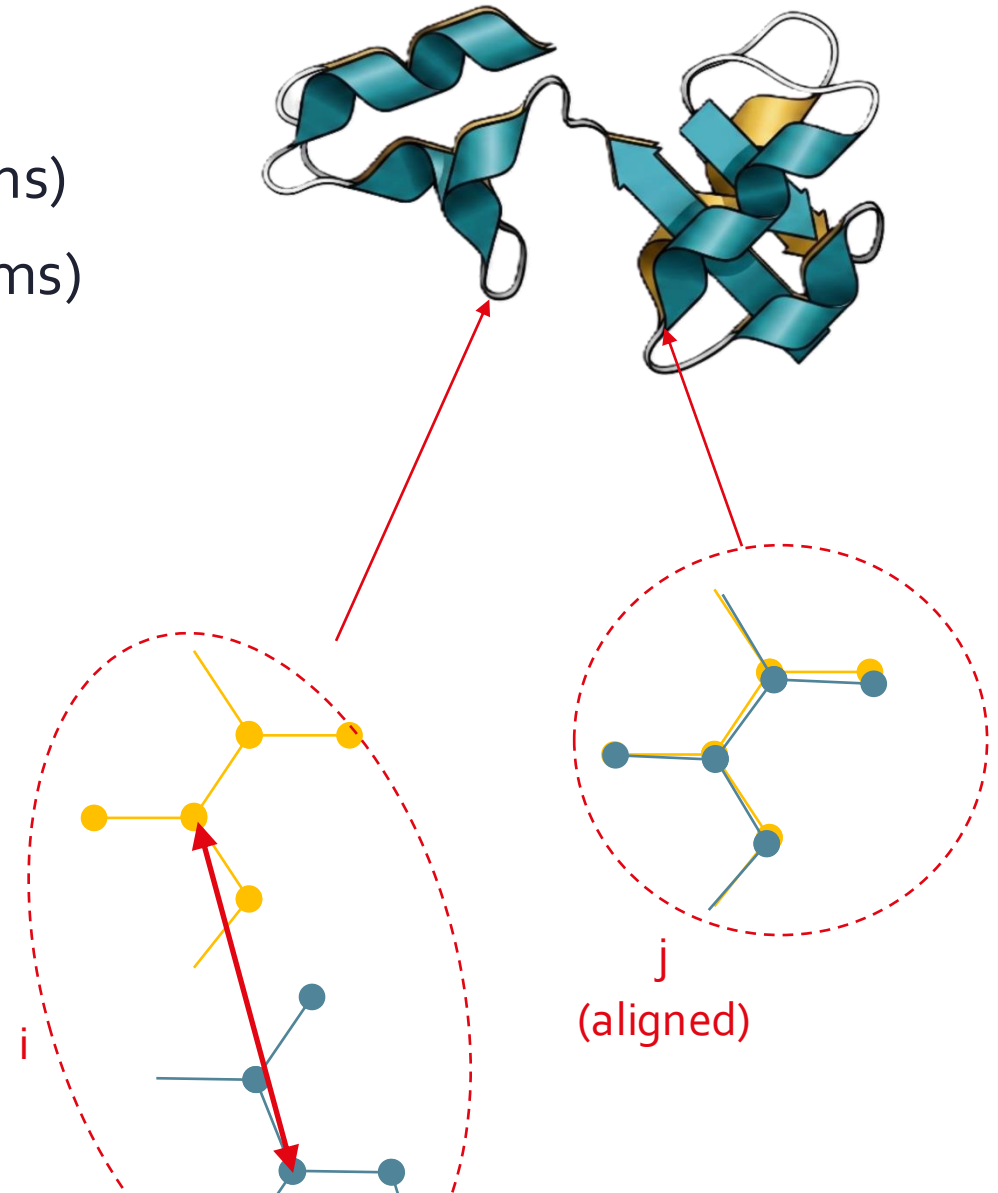
- For a pair of residues (i and j)
- Align around one residue (i) (N, C_{α} , C atoms)
- Difference at the other residue (j) (C_{α} atoms)
- Not symmetric





AE (aligned error)

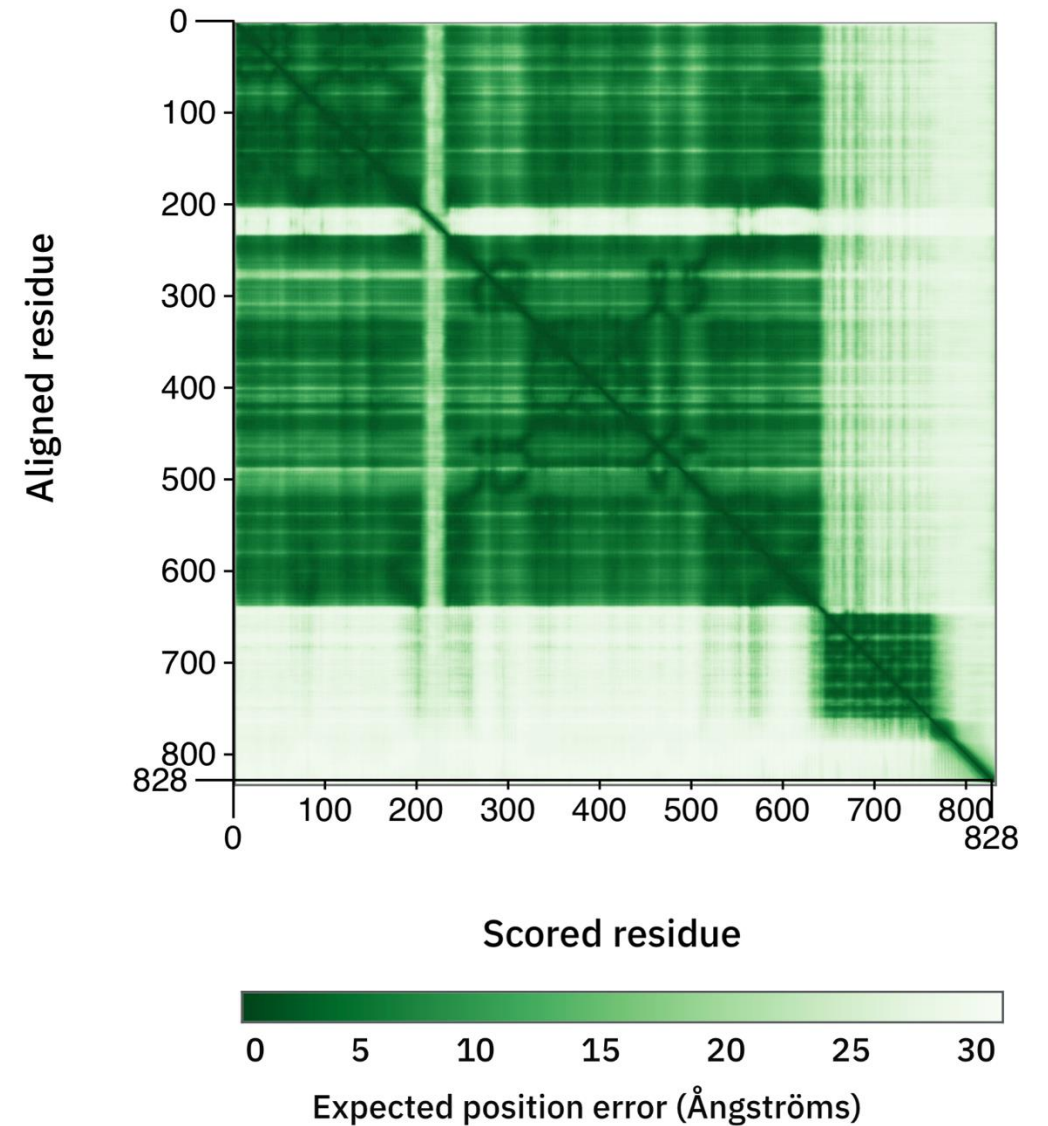
- For a pair of residues (i and j)
- Align around one residue (i) (N, C_{α} , C atoms)
- Difference at the other residue (j) (C_{α} atoms)
- Not symmetric





AE (aligned error)

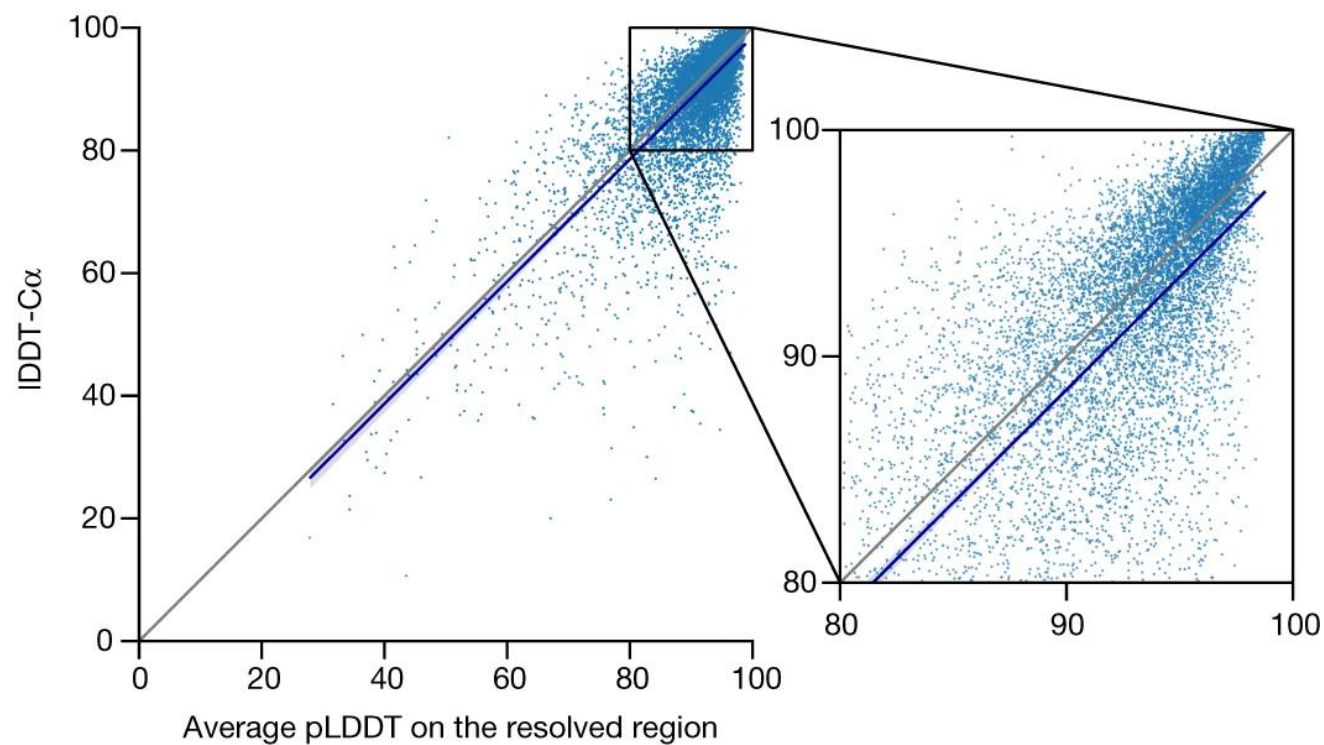
- For a pair of residues (i and j)
- Align around one residue (i)
- Difference at the other residue (j)
- Not symmetric



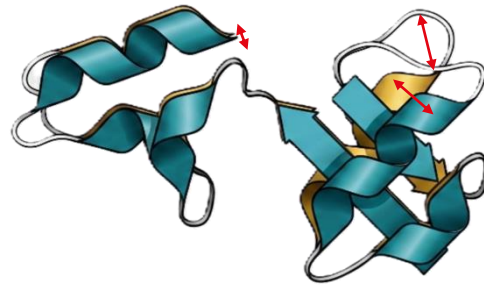


Confidence metrics

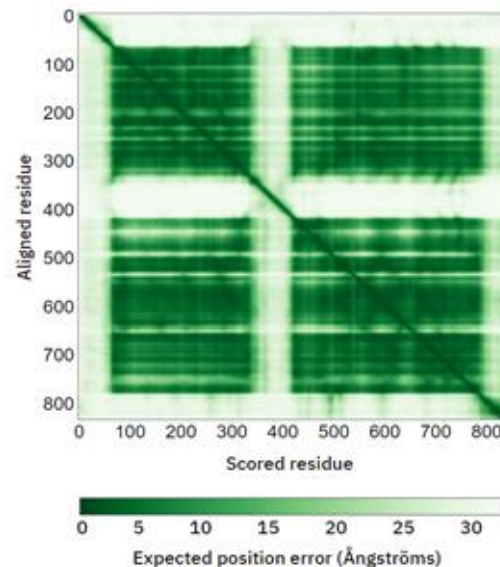
- Predicted
 - LDDT $\rightarrow$ pLDDT
 - AE $\rightarrow$ PAE
 - TM $\rightarrow$ pTM
- Confidence of the model in the prediction
- Correlate with prediction quality (but not perfect)



- Can be thought of as being calculated from PAE (but isn't exactly in reality)



$$TM = \max \left(\frac{1}{L_N} \sum_{i=1}^{L_T} \frac{1}{1 + \left(\frac{d_i}{d_0} \right)^2} \right)$$



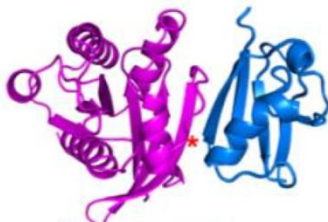
$$pTM = \max_i \left(\frac{1}{L} \sum_{j=1}^L \frac{1}{1 + \left(\frac{PAE_{ij}}{d_0} \right)^2} \right)$$



ipTM (interface pTM)

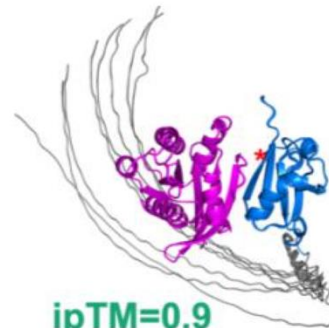
- pTM is calculated for the whole complex
- ipTM is calculated only for the interface
- Non-interacting domains and disordered regions can reduce it

Example 1:
no disorder in
either chain



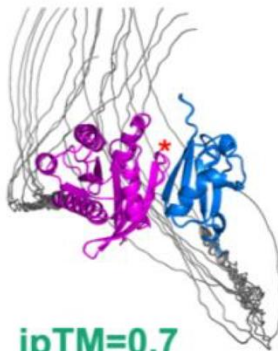
ipTM=0.9

Example 2:
60 disordered
residues in one
chain



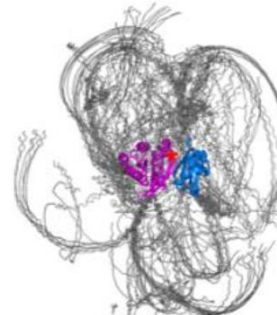
ipTM=0.9

Example 3:
60 residues
disorder in both
chains

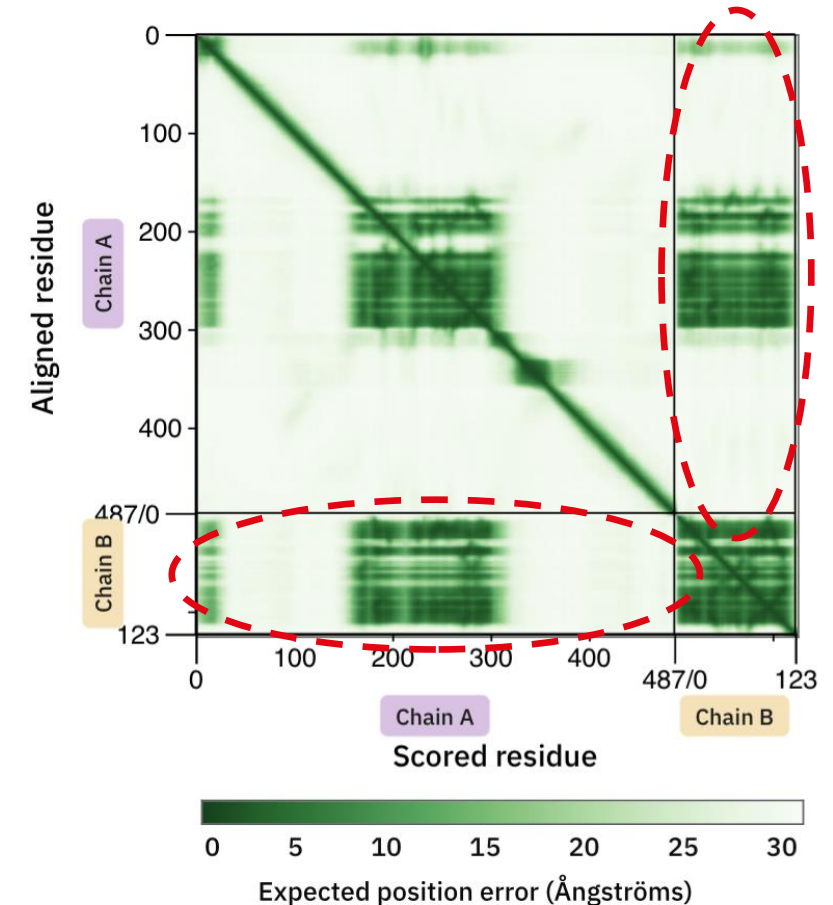


ipTM=0.7

Example 4:
120 disordered
residues in both
chains



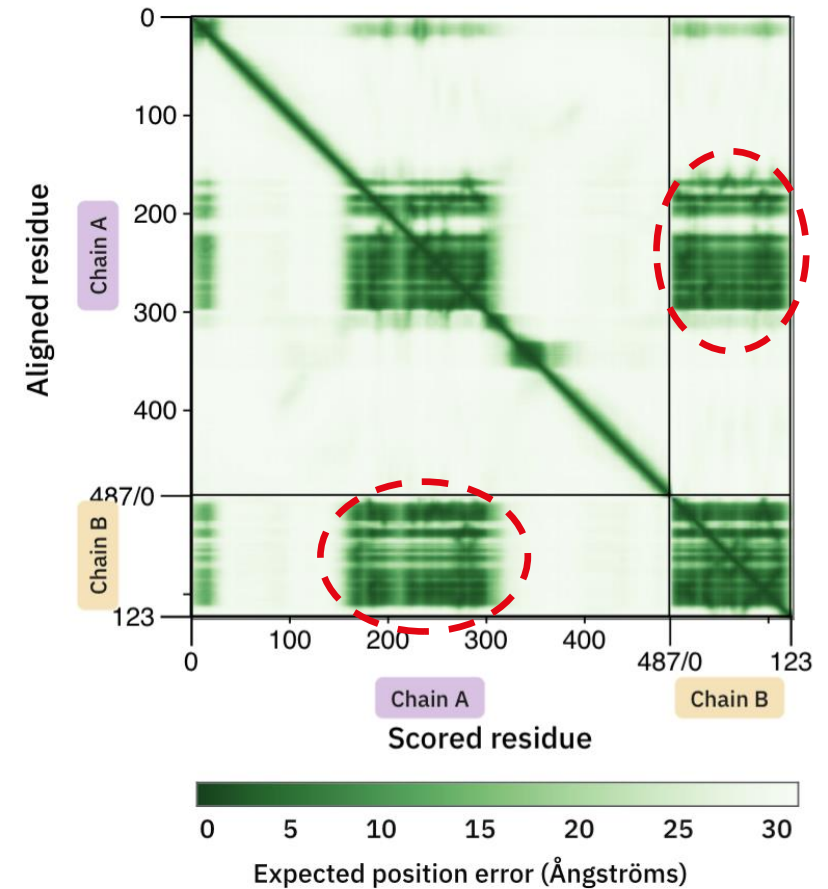
ipTM=0.6





ipSAE (interaction prediction Score from Aligned Errors)

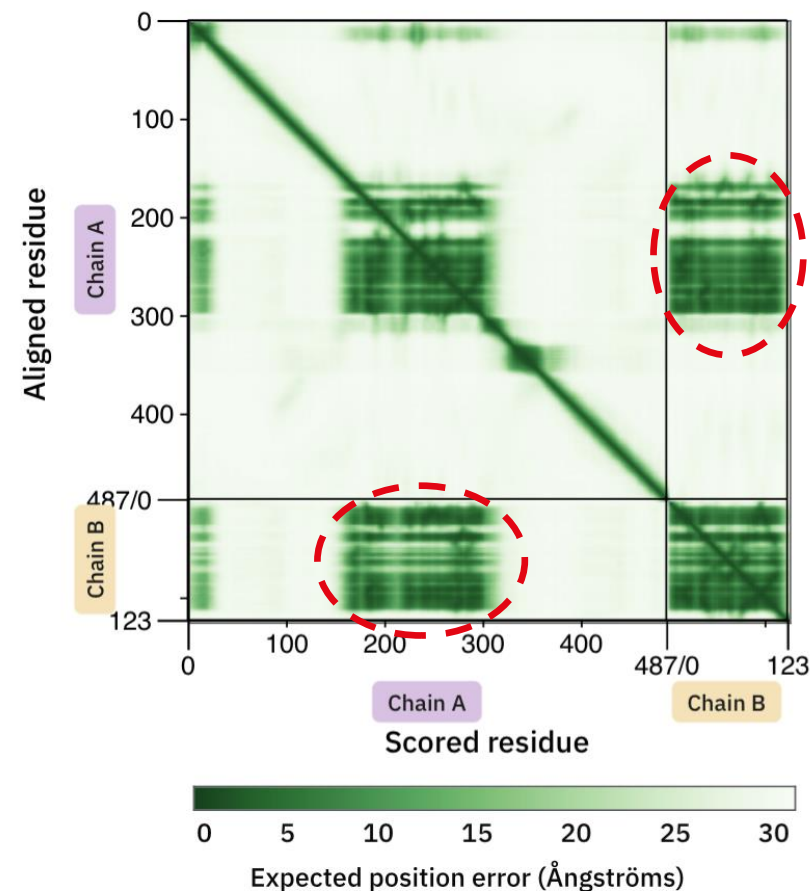
- Improved ipTM
- Includes only residue pairs with good PAE scores
 - If both interacting chains contain a lot of disorder or accessory contacts that don't form the contact, it ignores those
- Calculated from PAE
 - Not calculated by AlphaFold
 - Calculated separately, from the AlphaFold outputs





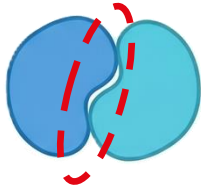
LIS (local interaction score)

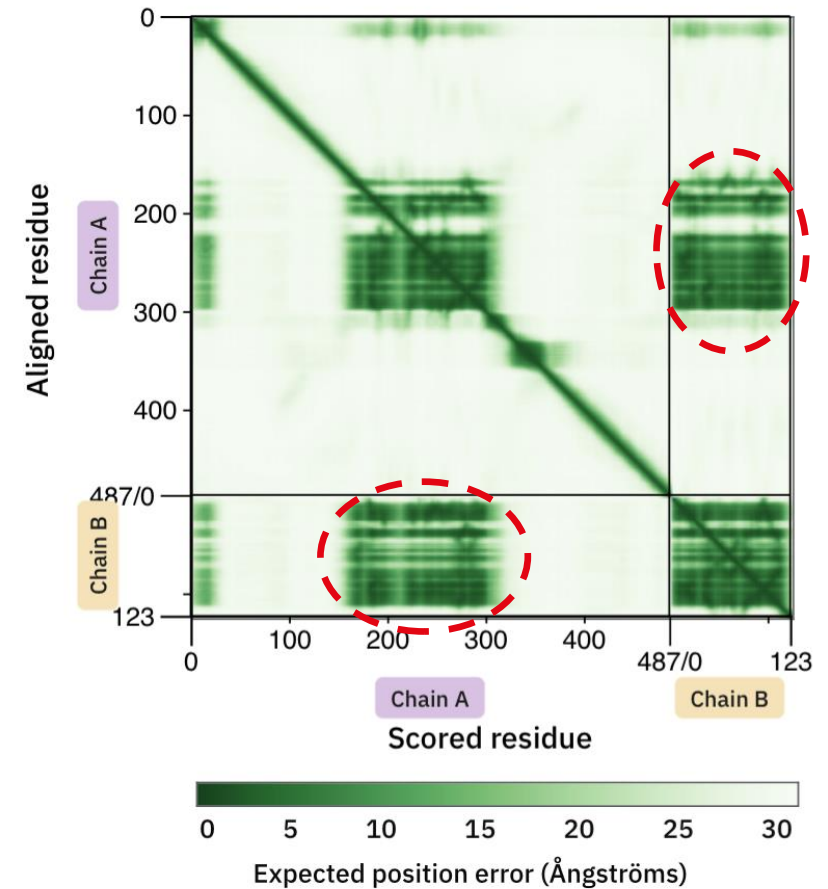
- conceptually similar to ipSAE
- Includes only residue pairs with good PAE scores
- Calculated from PAE
 - Not calculated by AlphaFold
 - Calculated separately, from the AlphaFold outputs





cLIS (local interaction score)

- Includes only residue pairs with good PAE scores
 - Includes only residue pairs that are physically close
- 
- Calculated from PAE
 - Not calculated by AlphaFold
 - Calculated separately, from the AlphaFold outputs





PAE Viewer

For interactive visualization of PAE matrix:

<https://pae-viewer.uni-goettingen.de/>

You can visualize dynamic PAE plots and match PAE data onto predicted structures stored in the **fold_<job_name>_model_<N>.cif** files

PAE VIEWER Interactive display of the Predicted Aligned Error

An error that occurs for some score JSON files which include the pLDDT has been fixed!

If you're experiencing any display issues, please make sure you're using the newest version of your browser. The webpage was tested on Windows 10 with version 108 of Firefox, Chrome and Edge. On iOS, version 111 of Firefox and Chrome was used, as well as Safari 16.3 and Opera 97.0.

Also, if the page is unresponsive, please make sure you're not using a browser plug-in blocking the cookie consent banner.

Examples Upload Offline version Citation

Choose one of the following examples or use the 'Upload' tab to upload your own protein complex data. The examples are complexes predicted by AlphaFold-Multimer from O'Reilly *et al.* (2023).

PdhA-PdhB-PdhI SufD-SufC-SufB FIIT-FIID

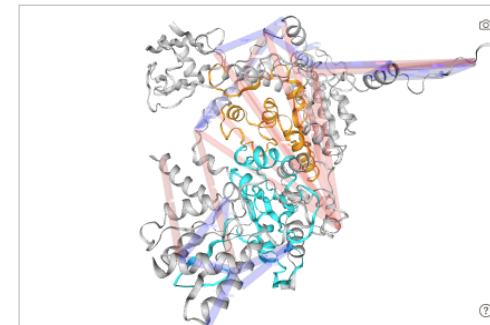
PdhA (371 aa)

MAAKTKKAIYFEEKQDAIEKKQFETFOILNEKGEVNEAAMPDLTDQGLKEIMRMVFTFVLDORSISLHGGRLGYAFIAGGASQIAERFALEKEQVLPGYADVTGSLWHGLFLYAFIFSRGHFSGVNPFDVNAISGQIIIGAGTGTAGVAIGLKKAGKAVAIITTDGGGASGGOFYEGINAGAYKAPAIWYQNNRYAISTYVEKQSAETIAQKAVAAGIVGVQDGMPLAVTAATAEKARAINGEGFTLETLTFRIGFHTMAGDOFTKYRKEIENEWKQDPLVFRAPLENKGLMSEEEKVIE
DAKEEIKQAIRKKADEAFQKVTDMKIMYERMPHNLSEGFETTTQRESK

PdhB (325 aa)

MAQMTIMQAITDALATELKNSENVLVFGEVGVNGGVFRADTEGLQKEFGSDRVFDTPLAESPIGGLALGSLNGFRFVMS4QFFGVVTEYBDSVSGQWAMRYASGGKWTSPVTIRSPFSGSVHTPELHAGSLEGLVAQSGIKVVISPTDAGLLISA
IRONDPVVFLRMKLYASFQGEVFEEETIELGKADVKREGTDLSTIITYGAMVHESLKAADLEKDGISAEVVDLRTVSLPDIOTIIASVEKTGRAIVVQEAQRQAGIAANVVAEINDRAILSLEAPVLRVAAPDTVFFFSQAESVWLPNHRKDVLETARKV
LEF

Selected: X: GLY 85 (PdhB) - GLY 211 (PdhB), Y: ARG 71 (PdhA) - ALA 179 (PdhA)



PdhA - PdhB - PdhI

Confidence metrics:

Mean pLDDT²: 95.4577

pTM²: 0.9229

ipTM²: 0.8940

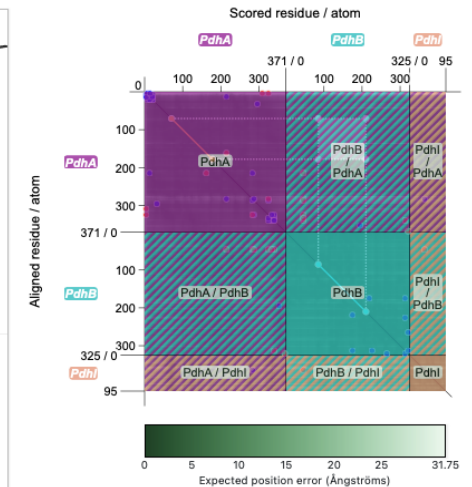
Structure color scheme:

- ☒ by subunit
- ☐ model confidence (pLDDT)

Crosslink color scheme:

- ☒ satisfied / ☐ violated restraint
- ☐ intra / ☐ inter protein links
- ☐ hide all crosslinks

Note: a Ca-Ca distance ≥ 30 Å is considered a restraint violation.



Cursor:

Selection:

X: GLY 85 (PdhB) - GLY 211 (PdhB), Y: ARG 71 (PdhA) - ALA 179 (PdhA)

toggle region overlay

(alternatively, hold the 'Shift' key when the cursor is within the PAE Viewer)

Download PAE matrix image

Download graph as SVG



Browser-based tools for analyzing protein-protein interactions from structure predictions. All analysis runs locally in your browser — no data leaves your device and no installation is needed.

Web: <https://flyark.github.io/LIVIA/>

GitHub: <https://github.com/flyark/LIVIA.git>

Preprint: [Kim & Perrimon 2026, *bioRxiv*](#)